

A03 Neural Network Zoo - Contributions

Nichelle:

For the Neural Network Zoo project I researched and developed the real-world applications section for Graph Neural Networks, focusing on why relational graph structure makes GNNs useful in fraud detection, recommendation systems, and molecular modeling. I added current scholarly references and translated the research into presentation-ready slide.

I also added a References and Contributions slide to the PowerPoint presentation for our team to cite sources and identify contributions.

I created a OneDrive file for our team to collaborate and collect our contributions, and compiled our parts. I created our GitHub repo to land our final submission(s) and use the link for our submission to Canvas.

Cesar:

I developed Section 1 (Introduction to Neural Networks) of the project, defining the essential theoretical foundations of deep learning and its key differences from classical machine learning.

I detailed the mathematical and structural mechanisms of the artificial neuron (Perceptron)—explaining the inputs (x_i), weights (w_i), bias (b), the summation equation $z = \sum (x_i \cdot w_i) + b$, and activation functions—while establishing the three-layer network architecture ($\text{Input} \rightarrow \text{Hidden Layers} \rightarrow \text{Output}$).

Additionally, I incorporated a transitional narrative to introduce the concept of the "Neural Network Zoo," thereby seamlessly connecting the introductory theory with our team's specialized focus on Graph Neural Networks (GNNs).

Mary Ann:

I researched how the Graphical Neural Network works and there was discussion as to how best align the animal with the network among the team. I decided to incorporate all the suggestions as to the chosen animal, as each had strengths and weaknesses, and also it would show how our thinking evolved.

I also put together some slides for the network portion of the presentation

Luiz:

For this group assignment, my individual contribution was researching the (GNN). My group chose the wolf as the animal to represent the GNN, so I also worked on connecting the idea of a wolf pack with how a GNN works.

I learned that a GNN uses a graph to represent information. The main parts of a graph are nodes and edges. Nodes represent objects, and edges represent the relationships or connections between the objects. For our wolf example, each wolf can represent a node, and the relationships between the wolves can represent the edges. Another important part I researched was message passing. This is how the nodes in a GNN share information with their neighbors. Each node receives information from the nodes connected to it and uses that information to update its own information. This process can happen through multiple GNN layers. I also learned that the number of layers can affect how much information a node can receive from the rest of the graph. With one layer, a node can learn from its direct neighbors. With more layers, information can travel farther through the graph.

For my part of the presentation, I helped create three slides about the GNN structure. The slides explain the nodes and edges, how information moves through the GNN layers, and how the wolf represents the GNN. I used the wolf pack as an example because wolves depend on their connections and communication with other members of the pack.

This research helped me understand GNNs better. Before working on this assignment, I did not know much about how graph-based neural networks worked. Now I understand that GNNs are focused not only on individual objects, but also on the relationships between those objects